Ratios and Equivalent Ratio Tables

Writing ratios in word, colon, and fraction form; part-to-part and part-to-whole; equivalent ratios

Directions:

- A ratio compares two quantities. You can write the same ratio three ways: with the word "to," with a colon, or as a fraction.
- Write ratios in simplest form when the problem asks for it, and show how you simplified in the work space.

1. The picture shows a group of counters. Count the shaded counters and the unshaded counters, then write the ratio of **shaded to unshaded** counters three ways.

word form: _________ colon form: _________ fraction form: _________

2. A fruit bowl holds 4 apples and 7 oranges. Write the ratio of apples to oranges two ways.

colon form: _________ fraction form: _________

3. The picture shows shaded and unshaded tiles. First find the **total** number of tiles, then write each ratio as a fraction.

(a) shaded tiles to **all** tiles: ________

(b) unshaded tiles to **all** tiles: ________

4. Are 6:8 and 9:12 equivalent ratios? Simplify each ratio in the work space, then write = or ≠ in the blank.

$$\frac{6}{8} \quad \underline{\hspace{2cm}} \quad \frac{9}{12}$$

5. Mr. Lee's class has 12 girls and 15 boys. Write the ratio of girls to boys as a fraction in **simplest form**.

Answer: ________________

6. A bag holds only red and blue marbles in the ratio 3:2 (red to blue). The bag holds 20 marbles in all. How many of the marbles are red?

Answer: ________________

7. Fill in the missing value that makes the two ratios equivalent:

$$\frac{5}{8} = \frac{?}{40}$$

Answer: ________________

8. A shelter cares for 6 dogs and 10 cats. Write each ratio as a fraction in simplest form.

(a) dogs to cats: ________

(b) cats to **all** animals: ________

9. Write these three ratios in order from **least to greatest**. It helps to rewrite each one as a fraction with the same denominator, or as a decimal.

5:8 3:4 7:10

least _________ _________ _________ greatest

10. In a class of 25 students, the ratio of students who walk to school to students who ride the bus is 2:3. Sam says, "The ratio starts with 2, so exactly 2 students walk." Find the number of students who actually walk, and explain in the work space why the 2 in the ratio does not mean exactly 2 students.

Answer: __________________